



Special Techniques in Electrostatics

Coulomb's Law (tedious in terms of vector nature & the evaluation of integral)

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\hat{r}}{r^2} \rho(\vec{r}') d\tau'$$

Gauss's Law (Only when symmetry permits)

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$$

Calculation of potential (to simplify the procedure)

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_P \frac{1}{r} \rho(\vec{r}') d\tau'$$

$$\vec{E} = -\vec{\nabla} V$$

Still, one needs to evaluate the tedious integral.

Further, if the conductors are involved, situation is more complicated as ρ itself may not be known in advance!



Special Techniques in Electrostatics

We need to find a simpler, general, accurate and powerful technique to calculate EF for any charge distribution

$$\boxed{\nabla^2 V = -\frac{\rho}{\epsilon_0}} \quad \text{Poisson's Equation}$$

In the region, where there is no charge,

$$\boxed{\nabla^2 V = 0} \quad \text{Laplace's Equation}$$

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0 \qquad \vec{\nabla} \times \vec{E} = 0 \qquad \vec{E} = -\vec{\nabla} V$$



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We need to find a simpler, general, accurate and powerful technique to calculate EF for any charge distribution

$$\nabla^2 V = -\frac{\rho}{\epsilon_0} \quad \text{Poisson's Equation}$$

$$\nabla^2 V = 0 \quad \text{Laplace's Equation}$$

PE/LE itself does not determine V .

Example:
$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

One dimension problem:
$$\frac{d^2 V}{dx^2} = 0 \quad y = mx + n$$

We need two appropriate boundary conditions to get the solution.



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$$\nabla^2 V = -\frac{\rho}{\epsilon_0} \quad \text{Poisson's Equation}$$

$$\nabla^2 V = 0 \quad \text{Laplace's Equation}$$

PE/LE itself does not determine V .

Example:
$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

Two-dimension problem:
$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

Things are very complicated here.

Problems in 3D are even more complicated!

Special Techniques in Electrostatics

We need to find a simpler, general, accurate and powerful technique to calculate EF for any charge distribution

$$\nabla^2 V = -\frac{\rho}{\epsilon_0} \quad \text{Poisson's Equation}$$

$$\nabla^2 V = 0 \quad \text{Laplace's Equation}$$

PE/LE itself does not determine V . **What we need in addition is an appropriate boundary conditions.**



What are those appropriate BCs, sufficient to determine V ?



Does PE/LE along with appropriate BC lead to the unique solution for the EF even in the presence of conductor?

YES

Uniqueness Theorem

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Uniqueness Theorem

$$\nabla \cdot \mathbf{E}_1 = \frac{1}{\epsilon_0} \rho \quad \nabla \cdot \mathbf{E}_2 = \frac{1}{\epsilon_0} \rho$$

$$\nabla \times \mathbf{E}_1 = 0 \quad \mathbf{E}_1 = -\nabla V_1 \quad \nabla \times \mathbf{E}_2 = 0 \quad \mathbf{E}_2 = -\nabla V_2$$

$$\mathbf{E}_3 \equiv \mathbf{E}_1 - \mathbf{E}_2,$$

$$\nabla \cdot \mathbf{E}_3 = 0 \quad \nabla \times \mathbf{E}_3 = 0 \quad \mathbf{E}_3 = -\nabla V_3 \quad V_3 \equiv V_1 - V_2$$

$$\nabla \cdot (V_3 \mathbf{E}_3) = V_3 (\nabla \cdot \mathbf{E}_3) + \mathbf{E}_3 \cdot (\nabla V_3) = -(E_3)^2.$$

$= 0$

$$\int_V \nabla \cdot (V_3 \mathbf{E}_3) d\tau = \oint_S V_3 \mathbf{E}_3 \cdot d\mathbf{a} = - \int_V (E_3)^2 d\tau$$

$$\int_V |\mathbf{E}|^2 d\tau = - \oint_S V \mathbf{E} \cdot d\mathbf{a}$$



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Answer: Uniqueness Theorem

$$\int_V |\vec{E}|^2 d\tau = -\oint_S V \vec{E} \cdot d\vec{a} \quad \vec{E} = \vec{E}_1 - \vec{E}_2 \quad V = V_1 - V_2$$

Any BC, which makes the surface integral zero, makes $\vec{E} = 0$ and hence the solution unique.

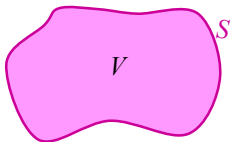
- Note:**
- 🍁 We do not know the region (V and S) of integration
 - 🍁 We do not know, what are specific BCs
 - 🍁 What we know is, BC should be such that the **Surface Integral must vanish**
 - 🍁 This only is sufficient to prove that the solution of PE/LE are unique!



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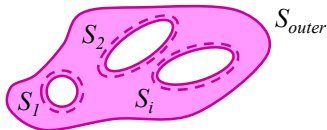
Region (S and V) of Integration:

A: In the absence of conductor but no cavity



S can be taken to infinity as well. Then integration is over all space and V generally taken to be zero there.

B: In the absence of conductor but having cavity

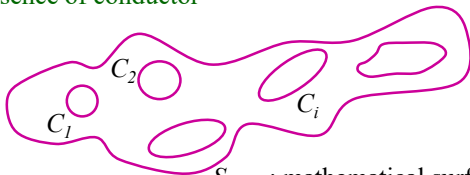


$$\int_V |\vec{E}|^2 d\tau = - \sum_i \oint_{S_i} V \vec{E} \cdot d\vec{a}$$

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Region (S and V) of Integration:

A: In the presence of conductor



S_{outer} : mathematical surface at infinity/finite surface enclosing the entire volume

$$\int_V |\mathbf{E}|^2 d\tau = - \sum_i \oint_{S_i} V \mathbf{E} \cdot d\vec{a}$$

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Boundary Conditions:

$$\int_V |\vec{E}|^2 d\tau = - \sum_i \oint_{S_i} V \vec{E} \cdot d\vec{a}$$



BC to make surface integration vanish is that V vanishes on all surfaces S_i

→ $V_1 = V_2 = \dots$ on each surface

→ **Potential is specified on the surface**

Dirichlet Boundary Conditions



BC to make surface integration vanish is that the normal component of $\vec{E}$ ($= \hat{n} \cdot \vec{E}$) vanishes on all surfaces S_i

→ $\vec{E}^\perp_1 = \vec{E}^\perp_2 = \dots$ on each surface

→ **Normal component of the EF is specified on the surface**

Neumann Boundary Conditions



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First Uniqueness Theorem of ED:

$$\int_V |\vec{E}|^2 d\tau = -\sum_i \oint_{S_i} V \vec{E} \cdot d\vec{a}$$

The EF in a volume V is **uniquely defined** by the solution of PE, if either

- the **potential**, or
- **EF's normal derivative** (directional derivative,
 $\vec{E}^\perp = \vec{E} \cdot \hat{n} = -\hat{n} \cdot \vec{\nabla} V$)

is specified on each surface of the volume.



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Second Uniqueness Theorem of ED (in the presence of conductor):

So far surface of volume V could be arbitrary surface on which BC was specified.

If any surface is the surface of a conductor, that must be an equipotential with constant value of V over the surface of the conductor.

$$\int_V |\vec{E}|^2 d\tau = - \sum_i V_i \oint_{S_i} \vec{E} \cdot d\vec{a}$$

Means, the normal component of $\vec{E}$ must vanish $\rightarrow \vec{E}^\perp_1 = \vec{E}^\perp_2 = \dots$ on each surface $\rightarrow$ Normal component of the EF is specified on the surface.

But, for conductor, $E^\perp = \sigma / \epsilon_0$. Thus, evaluation of $\oint_S \vec{E} \cdot d\vec{a}$ is equal to the net charge on the conductor.

EF in a volume V is uniquely defined by the solution of PE, if either the potential or the net charge is specified on each conductor.



Special Techniques in Electrostatics

Second Uniqueness Theorem of ED (in the presence of conductor):

If any surface is the surface of a conductor, that must be an equipotential with constant value of V over the surface of the conductor.

$$\int_V |\vec{E}|^2 d\tau = - \sum_i V_i \oint_{S_i} \vec{E} \cdot d\vec{a}$$

Means, the normal component of $\vec{E}$ must vanish $\rightarrow \vec{E}^\perp_1 = \vec{E}^\perp_2 = \dots$ on each surface $\rightarrow$ Normal component of the EF is specified on the surface.

But, for conductor $E^\perp = \sigma / \epsilon_0$. Thus, evaluation of $\oint_S \vec{E} \cdot d\vec{a}$ is equal to the net charge on the conductor.

EF in a volume V is uniquely defined by the solution of PE, if either the potential or the net charge is specified on each conductor.

Note: Actual surface charge distribution can not be known for a conductor until the problem is solved. Hence Neumann BC will not be a useful BC for conductors.

Consequence of Uniqueness Theorem of ED

Any method of solution is valid as long as the resulting potential satisfies PE/LE and the boundary conditions.

Even guessing a solution would be alright if it works.



Special Techniques in Electrostatics

Application (How to solve a problem)

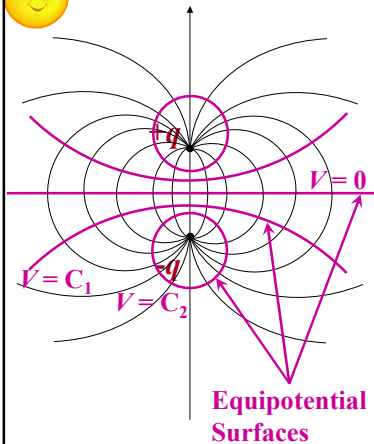
What we need...

- 🍁 Appropriate DE (PE/LE)
- 🍁 Region
- 🍁 Appropriate Boundary Conditions

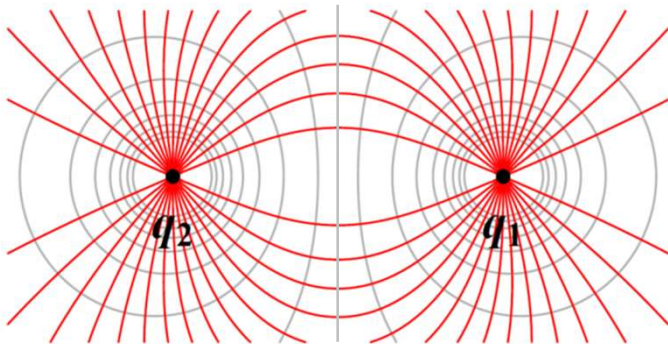
What we do...

- 🍁 Solve LE if $\rho = 0$, or PE if $\rho \neq 0$
 - Using direct integration if potential is a function of 1 variable
 - Using Separation of Variable if potential is a function of > 1 variable
- 🍁 Apply BC to determine unique solution
- 🍁 Having V , find the field using $\vec{E} = -\vec{\nabla}V$
- 🍁 If desired, calculate induced charge density on the conductor

Special Techniques in Electrostatics



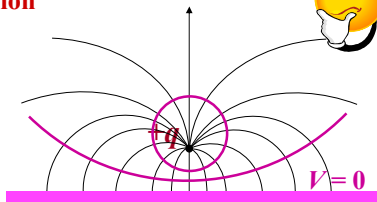
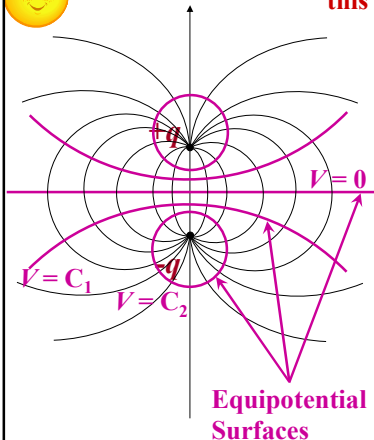
Special Techniques in Electrostatics



Special Techniques in Electrostatics



Nothing (E/V/charge distribution) will change in this region



Both the configuration will create the same field/potential in the region of observation

**Potential at a certain boundary is defined and is constant (0)
...which is the Boundary condition**



Special Techniques in Electrostatics



Nothing (E/V/charge distribution) will change in this region



$+q$ •

$+q$ •

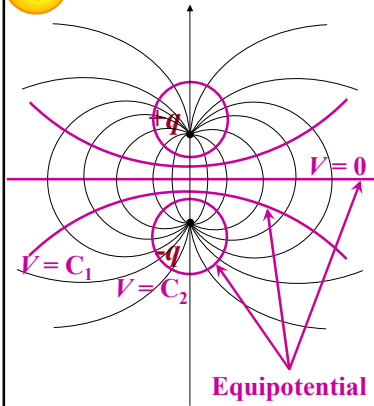
•
 $-q$

$V = 0$

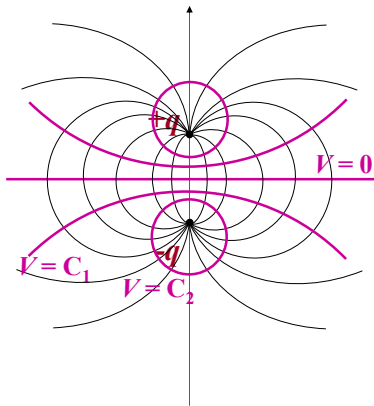


These two configurations are equivalent in the region of observation

Special Techniques in Electrostatics

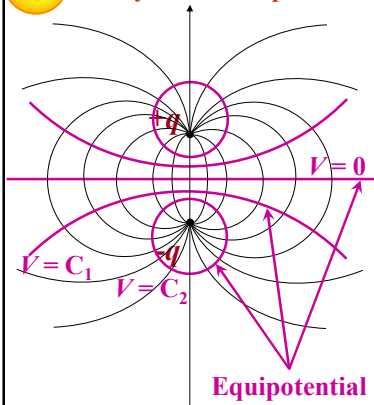


**Equipotential
Surfaces**

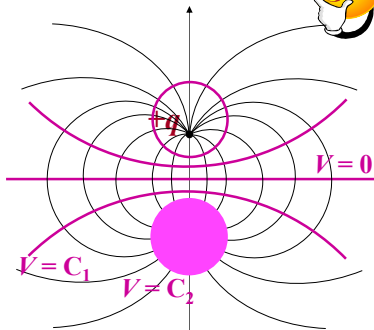


Special Techniques in Electrostatics

Nothing (E/V/charge distribution) will change anywhere except to the region within the conductor



Equipotential Surfaces

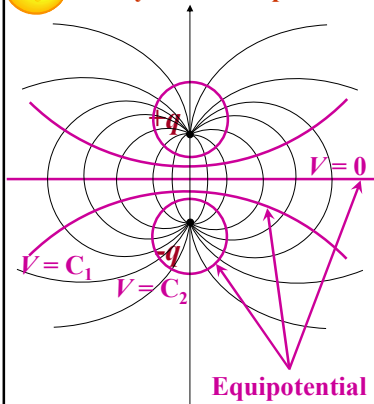


Both the configuration will create the same field/potential in the region of observation

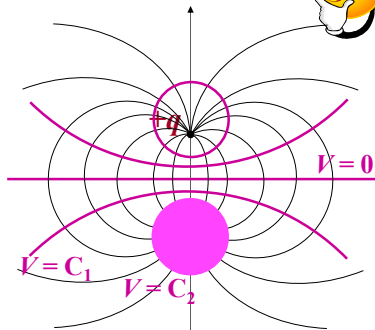
Special Techniques in Electrostatics



Nothing (E/V/charge distribution) will change anywhere except to the region within the conductor



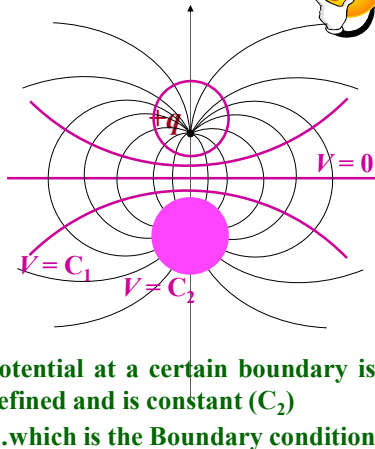
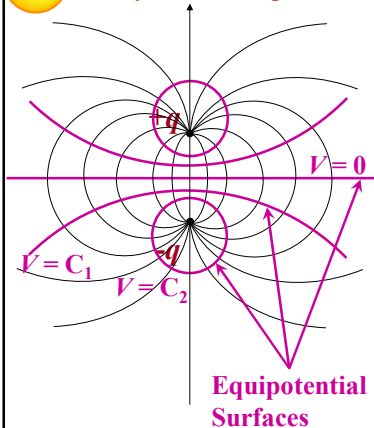
Equipotential Surfaces



Common point in both situation: potential at the boundary of sphere has to be strictly constant

Special Techniques in Electrostatics

Nothing (E/V/charge distribution) will change anywhere except to the region within the conductor



Special Techniques in Electrostatics



Nothing (E/V/charge distribution) will change anywhere except to the region within the conductor



$+q$ •

$+q$ •

•
 $-q$


 $V = C_2$

These two configurations are equivalent in the region of observation

Special Techniques in Electrostatics

What we have done is quite interesting!

$+q$ •

$+q$ •



$V = 0$



$V = C_2$

Find the EF/Potential from these configurations at a given point in the region above/outside the conductor

How to solve this problem?



Special Techniques in Electrostatics

What we have done is quite interesting!

$+q \bullet$

$V = 0$

Find the EF/Potential from these configurations at a given point in the region above/outside the conductor



How to solve this problem?

Is it $\frac{1}{4\pi\epsilon_0} \frac{q}{r}$? NO



Special Techniques in Electrostatics

What we have done is quite interesting!

$+q$ •

$+q$ •



$V = 0$



$V = C_2$

Find the EF/Potential from these configurations at a given point in the region above/outside the conductor



How to solve this problem?

Situation is more complicated here!

Special Techniques in Electrostatics

What we have done is quite interesting!

$+q$ •

$+q$ •



$V = 0$



$V = C_2$

Find the EF/Potential from these configurations at a given point in the region above/outside the conductor



How to solve this problem?

In both the cases, potential is defined at the conductor, which constitutes BC.

Special Techniques in Electrostatics

What we have done is quite interesting!



Find the EF/Potential from these configurations at a given point in the region above/outside the conductor

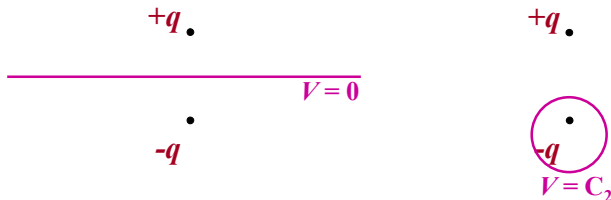


How to solve this problem?

Replace the given system with two equivalent point charges – one original, and another introduced at a proper location within conductor/conductor (image) plane

Special Techniques in Electrostatics

What we have done is quite interesting!



Find the EF/Potential from these configurations at a given point in the region above/outside the conductor



How to solve this problem?

only condition – point charge system should not alter the boundary potential and actual charge distribution in the region where the field/potential is desired



Special Techniques in Electrostatics

What we have done is quite interesting!

Summary

-  Remove the boundary (conductor defines the boundary), which is equipotential.
-  Introduce small number of point charge(s) (image charge) of **proper magnitude** at **appropriate location outside the region of interest**.
-  This must be done in a way that the individual scalar potential of all charges (**original** as well as **image** charges) must add up to **form an equipotential surface of the same shape and same value** as the **original boundary surface**.



METHOD OF IMAGES



Special Techniques in Electrostatics

METHOD OF IMAGES

Always remember

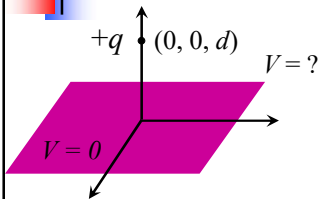


Image charges must be located in the conducting region (outside the region of interest). **IS A MUST** – as this condition is necessary to satisfy Poisson Equation.



Image charges must be located such that on the conducting surface, potential is zero/constant. **IS A MUST** – as this condition ensures that BCs are satisfied.

METHOD OF IMAGES: Example 1



Infinite grounded conducting plane

**What we should actually
do is like the following:**

As charge is given & potential
at a given surface is defined
(zero here)

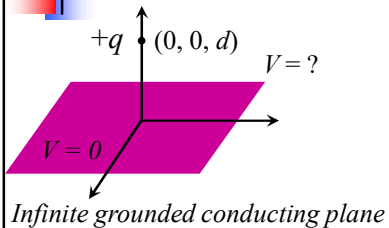
We need to solve PE in a region $z > 0$ with a single point charge at $(0, 0, d)$ & apply the BC as:

- (i) $V = 0$ at $z = 0$ plane
- (ii) $V \rightarrow 0$ far away from the charge ($x^2 + y^2 + z^2 \gg d^2$)

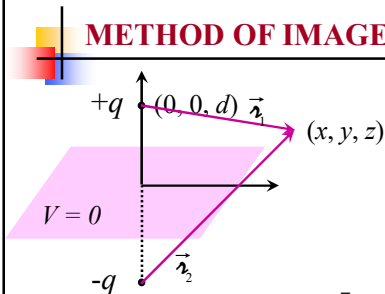
Uniqueness Theorem guarantees that there will be only one function that meets this requirement

Even by making a guess, if we can discover that function, it would be the right answer!

METHOD OF IMAGES: Example 1



METHOD OF IMAGES: Example 1



Potential Calculation

$$\vec{r}_1 = \sqrt{x^2 + y^2 + (z - d)^2}$$

$$\vec{r}_2 = \sqrt{x^2 + y^2 + (z + d)^2}$$

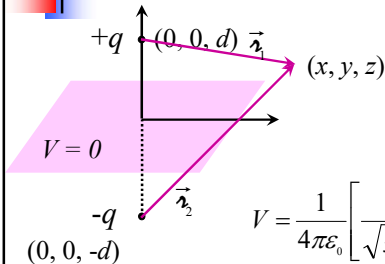
$$\begin{aligned} (0, 0, -d) \quad V = V_+ - V_- &= \frac{1}{4\pi\epsilon_0} \left[\frac{q}{r_1} - \frac{q}{r_2} \right] \\ &= \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + (z - d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z + d)^2}} \right] \end{aligned}$$

As can be observed from the above function:

- (i) $V = 0$ when $z = 0$
- (ii) $V \rightarrow 0$ when $(x^2 + y^2 + z^2 \gg d^2)$
- (iii) only charge in the region $z > 0$ is q at $(0, 0, d)$

**BCs are satisfied &
this is the only solution**

METHOD OF IMAGES: Example 1



Induced Charge Density

$$\sigma = -\epsilon_0 \frac{\partial V}{\partial n}$$

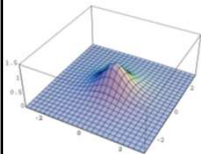
$$\sigma = -\epsilon_0 \left. \frac{\partial V}{\partial z} \right|_{z=0}$$

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + (z-d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z+d)^2}} \right]$$

$$\frac{\partial V}{\partial z} = \frac{1}{4\pi\epsilon_0} \left[\frac{-q(z-d)}{[x^2 + y^2 + (z-d)^2]^{3/2}} + \frac{q(z+d)}{[x^2 + y^2 + (z+d)^2]^{3/2}} \right]$$

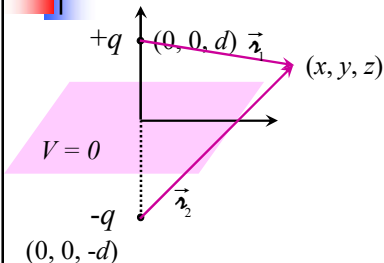
$$\therefore \sigma = \frac{-qd}{2\pi[x^2 + y^2 + d^2]^{3/2}}$$

Home-Work: Plot it for $x=0$, and then for $y=0$



Note: Its not a uniform charge density. Maximum at $x = y = z = 0$!

METHOD OF IMAGES: Example 1



As expected!

Total Induced Charge

$$\sigma = \frac{-qd}{2\pi[x^2 + y^2 + d^2]^{3/2}}$$

$$Q = \int \sigma da$$

$$= \iint \frac{-qd dx dy}{2\pi(x^2 + y^2 + d^2)^{3/2}}$$

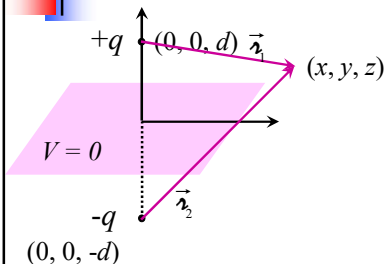
$$= \int_{r=0}^{\infty} \int_0^{2\pi} \frac{-qd r dr d\phi}{2\pi(r^2 + d^2)^{3/2}}$$

$$= \frac{-qd}{2} \int_{r=0}^{\infty} \frac{2r dr}{(r^2 + d^2)^{3/2}}$$

$$= qd \left[\frac{1}{\infty} - \frac{1}{d} \right] = -q$$

METHOD OF IMAGES: Example 1

Force



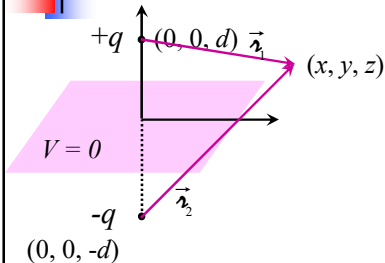
Charge q would be attracted towards the plane because of the induced $-ve$ charge.

This force would be same as the force experience by q due to $-q$.

$$\vec{F} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2d)^2} \hat{z}$$

METHOD OF IMAGES: Example 1

Energy



With two point charges ($q, -q$) and no conductor, it would be

$$W = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{2d}$$

Let us calculate energy more rigorously.

$$W = \int_{\infty}^d \vec{F} \cdot d\vec{l} = \frac{1}{4\pi\epsilon_0} \int_{\infty}^d \frac{q^2}{4z^2} \hat{z} \cdot dz \hat{z} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d}$$

$$\vec{F} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2d)^2} \hat{z}$$

Half of what METHOD OF IMAGES suggests!

What's wrong?

Nothing

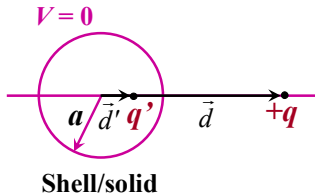
METHOD OF IMAGES: Example 2

BC: Potential at the conducting surface is zero.

Image charge:

Must be in conductor's plane, i.e., inside the conductor.

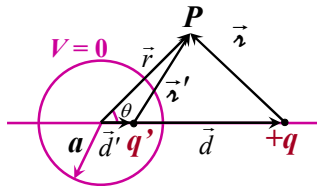
Magnitude and location (q' , d') of the image charge must be chosen such that the spherical surface at $r = a$ (in the absence of the conductor) becomes an equipotential with $V = 0$.



METHOD OF IMAGES: Example 2

Potential at P :

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|\vec{r} - \vec{d}|} + \frac{q'}{|\vec{r} - \vec{d}'|} \right] \quad r > a$$



Applying BC at $r = a$

$$\begin{aligned} V(\vec{r})|_{r=a} &= \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|a\hat{r} - d\hat{d}|} + \frac{q'}{|a\hat{r} - d'\hat{d}|} \right] = 0 \\ &= \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|a\hat{r} - d\hat{d}|} + \frac{q'}{\frac{a}{d} \left| d\hat{r} - \frac{dd'}{a} \hat{d} \right|} \right] = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|a\hat{r} - d\hat{d}|} + \frac{(d/a)q'}{\left| d\hat{r} - \frac{dd'}{a} \hat{d} \right|} \right] = 0 \end{aligned}$$

If $d' = a^2 / d$

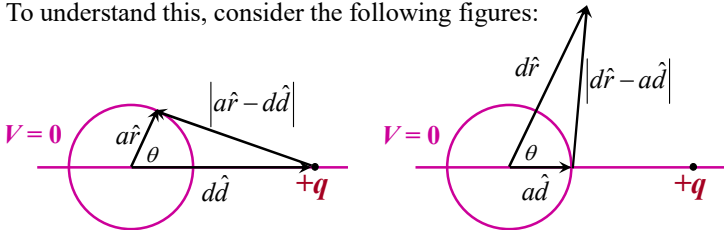
$$V(\vec{r})|_{r=a} = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|a\hat{r} - d\hat{d}|} + \frac{(d/a)q'}{|d\hat{r} - a\hat{d}|} \right]$$

Here, always $|a\hat{r} - d\hat{d}| = |d\hat{r} - a\hat{d}|$

METHOD OF IMAGES: Example 2

$$V(\vec{r})|_{r=a} = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|a\hat{r} - d\hat{d}|} + \frac{(d/a)q'}{|d\hat{r} - a\hat{d}|} \right] \quad \text{Here, always } |a\hat{r} - d\hat{d}| = |d\hat{r} - a\hat{d}|$$

To understand this, consider the following figures:



Using cosine law: $|a\hat{r} - d\hat{d}| = \sqrt{a^2 + d^2 - 2ad \cos \theta}$

$$|d\hat{r} - a\hat{d}| = \sqrt{d^2 + a^2 - 2da \cos \theta}$$



$|a\hat{r} - d\hat{d}|$ will always be equal to $|d\hat{r} - a\hat{d}|$

METHOD OF IMAGES: Example 2

Potential at P :

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|\vec{r} - \vec{d}|} + \frac{q'}{|\vec{r} - \vec{d}'|} \right] \quad r > a$$

Applying BC at $r = a$

$$V(\vec{r})|_{r=a} = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|a\hat{r} - d\hat{d}|} + \frac{q'}{|a\hat{r} - d'\hat{d}|} \right] = 0$$

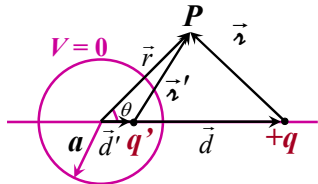
$$= \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|a\hat{r} - d\hat{d}|} + \frac{q'}{\frac{a}{d} |d\hat{r} - \frac{dd'}{a} \hat{d}|} \right] = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|a\hat{r} - d\hat{d}|} + \frac{(d/a)q'}{|d\hat{r} - \frac{dd'}{a} \hat{d}|} \right] = 0$$

If $d' = a^2 / d$

$$V(\vec{r})|_{r=a} = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|a\hat{r} - d\hat{d}|} + \frac{(d/a)q'}{|d\hat{r} - a\hat{d}|} \right]$$

This will be zero iff $q = -\left(\frac{d}{a}\right)q'$

$$q' = -\left(\frac{a}{d}\right)q \quad d' = \frac{a^2}{d}$$



METHOD OF IMAGES: Example 2

Potential at P :

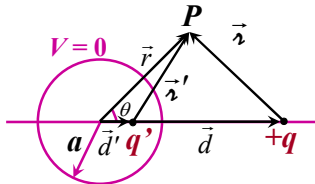
$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|\vec{r} - \vec{d}|} + \frac{q'}{|\vec{r} - \vec{d}'|} \right] \quad r > a$$

$$q' = -\left(\frac{a}{d}\right)q$$

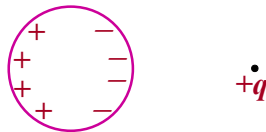
$$d' = \frac{a^2}{d}$$

We can now calculate E , F , q , and σ .

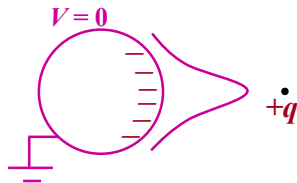
$$\vec{F} \text{ (on } q \text{ by an image charge)} = \frac{qq'}{4\pi\epsilon_0(d-d')^2} = \frac{-aq^2\vec{d}}{4\pi\epsilon_0(d^2 - a^2)^2}$$



METHOD OF IMAGES: Example 2



METHOD OF IMAGES: Example 2

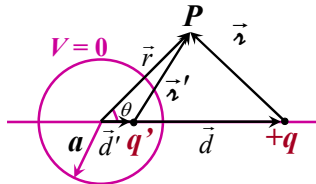


METHOD OF IMAGES

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|\vec{r} - \vec{d}|} + \frac{q'}{|\vec{r} - \vec{d}'|} \right] \quad r > a$$

$$q' = -\left(\frac{a}{d}\right)q$$

$$d' = \frac{a^2}{d}$$



The solution for the point charge outside a grounded sphere can be used as a starting point to find the solution of several related problems

METHOD OF IMAGES: Example 3

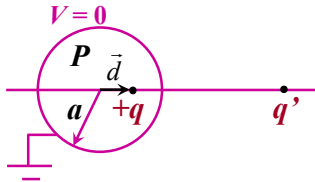
A point charge is kept inside a grounded hollow conducting sphere. Find potential within the sphere?

Solution will be same as before.

Cause, the solution of the previous problem never demanded $d > a$!

Image charge will now be outside the sphere

Solution will be valid now within the space of the hollow sphere



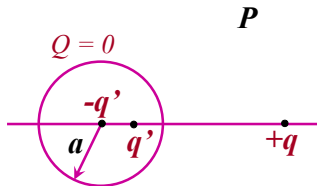
METHOD OF IMAGES: Example 4

A point charge outside an isolated, neutral conducting sphere.

BC: Net charge on the conducting surface is defined.

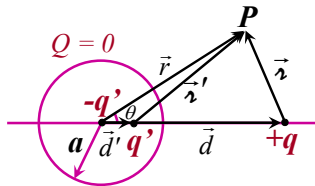
Starting point: Example 2.

- Start with a grounded conducting sphere in front of q . q' charge would be distributed over its surface.
- Disconnect the grounded wire.
- Add to the sphere a charge $-q'$ to make total charge zero and the conductor neutral.
- This added charge $-q'$ will distribute itself uniformly over the surface as the electrostatic force due to the point charge q is already balanced by the charge q' .
- Potential due to the added charge $-q'$ will be the same as if a point charge of this magnitude is kept at the origin



METHOD OF IMAGES: Example 4

A point charge outside an isolated, neutral conducting sphere.



$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|\vec{r} - \vec{d}|} + \frac{q'}{|\vec{r} - \vec{d}'|} + \frac{-q'}{|\vec{r}|} \right] \quad r > a$$

$$q' = -\left(\frac{a}{d}\right)q$$

$$d' = \frac{a^2}{d}$$

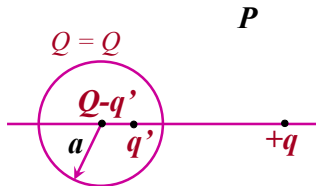
METHOD OF IMAGES: Example 5

A point charge outside an isolated, charged conducting sphere.

BC: Net charge on the conducting surface is defined.

Starting point: Example 2.

- Start with a grounded conducting sphere in front of q . q' charge would be distributed over its surface.
- Disconnect the grounded wire.
- Add to the sphere a charge $Q - q'$ to make total charge zero and the conductor neutral.
- This added charge $Q - q'$ will distribute itself uniformly over the surface as the electrostatic force due to the point charge q is already balanced by the charge q' .
- Potential due to the added charge $Q - q'$ will be the same as if a point charge of this magnitude is kept at the origin

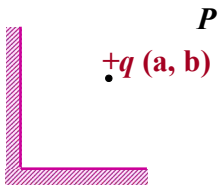




METHOD OF IMAGES: Example 6

Till now we have encountered only one conductor surface as the boundary for the problem.

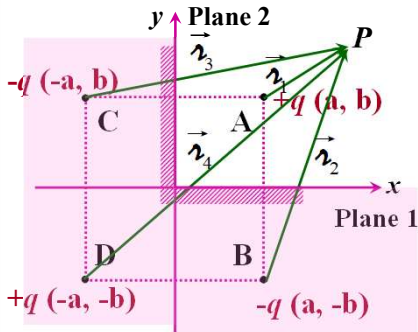
Consider a point charge in front of two semi-infinite grounded conductor planes at right angle. Find V in the region between the two conductors where charge is located.



METHOD OF IMAGES: Example 6

Till now we have encountered only one conductor surface as the boundary for the problem.

Consider a point charge in front of two semi-infinite grounded conductor planes at right angle. Find V in the region between the two conductors where charge is located.



$$V = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_1} - \frac{1}{r_2} - \frac{1}{r_3} + \frac{1}{r_4} \right]$$

$$r_1 = [(x-a)^2 + (y-b)^2 + z^2]^{1/2}$$

$$r_2 = [(x-a)^2 + (y+b)^2 + z^2]^{1/2}$$

$$r_3 = [(x+a)^2 + (y-b)^2 + z^2]^{1/2}$$

$$r_4 = [(x+a)^2 + (y+b)^2 + z^2]^{1/2}$$

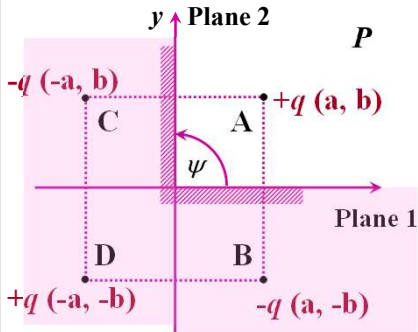
METHOD OF IMAGES: Example 6

Consider a point charge in front of two semi-infinite inclined grounded conductor planes. Find V in the region between the two conductors where charge is located.

Important Note

If the angle of inclination ψ is not an integer divisor of 180° , image charges can not satisfy the BC ($V=0$ at conductor)

In this case image charge will not remain only in the conducting/image plane, but will be in the region of interest.



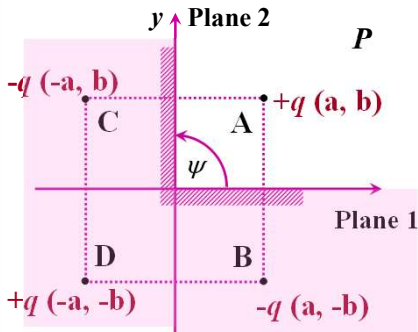
METHOD OF IMAGES: Example 6

Consider a point charge in front of two semi-infinite inclined grounded conductor planes. Find V in the region between the two conductors where charge is located.

Important Note

For the angle of inclination ψ being an integer divisor of 180° , total number of images would be

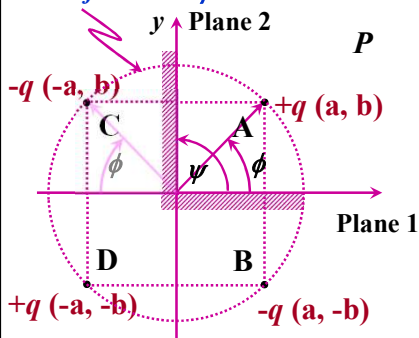
$$N = \left(\frac{360^\circ}{\psi} - 1 \right)$$



METHOD OF IMAGES: Example 6

Consider a point charge in front of two semi-infinite inclined grounded conductor planes. Find V in the region between the two conductors where charge is located.

Image charges on the circle of constant ρ



Important Note

Consider CPCS with z-axis along the line of intersection of the conductor surfaces.

Coordinates of q would be (ρ, ϕ) .

Reflection through Plane 1:

$$q(\rho, \phi) \rightarrow -q(\rho, -\phi)$$

Reflection through Plane 2:

$$q(\rho, \phi) \rightarrow -q(\rho, 2\psi - \phi)$$

Special Techniques in Electrostatics

Problem of first kind:

Surface of the conductor defined the BC and the region of interest where field was examined contained a point charge

Problem of second kind

Potential/charge density is specified on the boundary of some region, and we are asked to find potential/field in the interior where there is no point charge

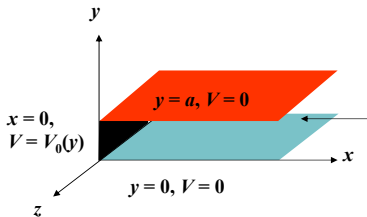
Almost in all such cases, boundary (conductor) surface have shapes that corresponds to constant coordinate surfaces in some CS

We can not use Method of Images, rather need to solve Laplace Equation!

Special Techniques in Electrostatics

Method needed to solve Laplace Eq is known as
SEPARATION OF VARIABLES

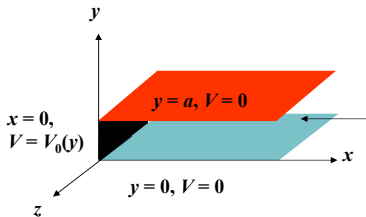
Example: 1



What is V here?

Special Techniques in Electrostatics: Example 1

Method needed to solve Laplace Eq is known as
SEPARATION OF VARIABLES



What is V here?

1. Look for symmetry

There is no charge in the region

Potential varies along x as well as along y

Potential does not vary along z at all



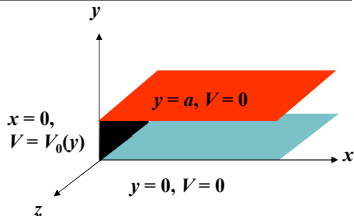
$$V = V(x, y)$$

(1)

Special Techniques in Electrostatics: Example 1

2. Define BC

- i. $V = 0$, at $y = 0$
- ii. $V = 0$, at $y = a$
- iii. $V = V_0(y)$ at $x = 0$
- iv. In addition: $V \rightarrow 0$ as $x \rightarrow \infty$ (not given but obvious)



3. Laplace Eq (Homogeneous DE)

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0 \quad (2)$$

4. Variable separable method

$$V(x, y) = X(x)Y(y) \quad (3)$$

~~$$V(x, y) = 6x + 10y$$~~

Special Techniques in Electrostatics: Example 1

Laplace Eq becomes

$$Y \frac{d^2 X}{dx^2} + X \frac{d^2 Y}{dy^2} = 0 \quad (4)$$

Dividing by (3) $V(x, y) = X(x)Y(y)$

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} = 0 \quad (5)$$

$$f(x) + g(y) = 0$$

$$C_1 + C_2 = 0$$

$$\longrightarrow \frac{1}{X} \frac{d^2 X}{dx^2} = C_1 \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = C_2 \quad \text{PDE} \rightarrow \text{ODE}$$

Special Techniques in Electrostatics: Example 1

$$\frac{1}{X} \frac{d^2 X}{dx^2} = C_1 \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = C_2 \quad \text{PDE} \rightarrow \text{ODE}$$

Assume $C_2 = +ve$

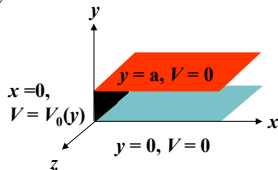
$$C_2 = k^2$$

$$\frac{1}{Y} \frac{d^2 Y}{dy^2} = k^2 \quad \longrightarrow \quad \frac{d^2 Y}{dy^2} - k^2 Y = 0$$

Solution would be $e^{\pm ky}$

$\longrightarrow$ Hence $C_1 = k^2 \quad C_2 = -k^2$

$$\frac{d^2 X}{dx^2} - k^2 X = 0 \quad \frac{d^2 Y}{dy^2} + k^2 Y = 0$$



Special Techniques in Electrostatics: Example 1

$$\frac{d^2 X}{dx^2} - k^2 X = 0 \qquad \frac{d^2 Y}{dy^2} + k^2 Y = 0 \qquad (6)$$

→ $X(x) = Ae^{kx} + Be^{-kx}$
 $Y(y) = C \sin ky + D \cos ky$

OR $V(x, y) = (Ae^{kx} + Be^{-kx}) (C \sin ky + D \cos ky) \qquad (7)$

Variable separable solution to the Laplace Equation

4. Application of Boundary Condition

iv. $V \rightarrow 0$ as $x \rightarrow \infty$ → $A \rightarrow 0$ Assuming k is +ve

$$V(x, y) = Be^{-kx} (C \sin ky + D \cos ky)$$

OR $V(x, y) = e^{-kx} (C \sin ky + D \cos ky)$

Special Techniques in Electrostatics: Example 1

4. Application of Boundary Condition

iv. $V \rightarrow 0$ as $x \rightarrow \infty$ $\longrightarrow$ $A \rightarrow 0$

$$V(x, y) = e^{-kx} (C \sin ky + D \cos ky)$$

i. $V \rightarrow 0$ at $y \rightarrow 0$ $\longrightarrow$ $D \rightarrow 0$

$$V(x, y) = Ce^{-kx} \sin ky$$

ii. $V \rightarrow 0$ at $y \rightarrow a$ $\longrightarrow$ $0 = Ce^{-kx} \sin ka$

$$k = n\pi/a \quad n = 1, 2, 3 \dots$$

Note: k can not be -ve and can not be zero

Observe: BCs are applied in a way to first make all possible constants zero (in this case A and D).

Special Techniques in Electrostatics: Example 1

Solution to LE for this problem

$$V(x, y) = Ce^{-n\pi x/a} \sin(n\pi y/a) \quad n = 1, 2, 3 \dots$$

Infinite Set

General Solution to LE , which is Homogeneous DE, for this problem

$$V(x, y) = \sum_{n=1}^{\infty} C_n e^{-n\pi x/a} \sin(n\pi y/a) \quad (8)$$

Application of Boundary Condition

iii. $V \rightarrow V_0(y)$ at $x \rightarrow 0$

$$V(0, y) = \sum_{n=1}^{\infty} C_n \sin(n\pi y/a) = V_0(y)$$

Special Techniques in Electrostatics: Example 1

Consider an interval $[a, b]$ for variable x

Consider functions $U_n(x)$:

$U_1(x), U_2(x), \dots$ defined on $[a, b]$

$$\int_a^b U_n(x) U_m(x) dx = s_n \delta_{mn} \quad \text{Orthogonal functions}$$

if $s_n = 1$ **Orthonormal functions**

$$\int_a^b U_n(x) U_m(x) dx = \begin{cases} 0 & m \neq n \\ 1 & m = n \end{cases}$$

$U_n(x): \{U_1(x), U_2(x), \dots\}$ form a complete set on $[a, b]$

$$\longrightarrow V(x) = \sum_{n=1}^{\infty} C_n U_n(x)$$

Special Techniques in Electrostatics: Example 1

Example: $\sin\left(\frac{n\pi x}{a}\right)$ in an interval $[0, a]$, n a +ve integer

$$\int_0^a \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi x}{a}\right) dx = \frac{a}{2} \delta_{mn}$$

→ **Orthogonal functions**

$\sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$ in an interval $[0, a]$, n a +ve integer

→ **Orthonormal functions**

We shall exploit this property of sin function

Special Techniques in Electrostatics: Example 1

$$V(0, y) = \sum_{n=1}^{\infty} C_n \sin(n\pi y / a) = V_0(y)$$

$$\int_0^a \sum_{n=1}^{\infty} C_n \sin(n\pi y / a) \sin(n'\pi y / a) dy = \int_0^a V_0(y) \sin(n'\pi y / a) dy$$

$$\sum_{n=1}^{\infty} C_n \int_0^a \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{n'\pi y}{a}\right) dy = \int_0^a V_0(y) \sin(n'\pi y / a) dy$$

$$\sum_{n=1}^{\infty} C_n \frac{a}{2} \delta_{nn'} = \int_0^a V_0(y) \sin(n'\pi y / a) dy$$

$$C_{n'} = \frac{2}{a} \int_0^a V_0(y) \sin(n'\pi y / a) dy$$

Special Techniques in Electrostatics: Example 1

Unique solution to the given problem:

$$V(x, y) = \sum_{n=1}^{\infty} C_n e^{-n\pi x/a} \sin(n\pi y/a)$$

$$C_{n'} = \frac{2}{a} \int_0^a V_0(y) \sin(n'\pi y/a) dy$$

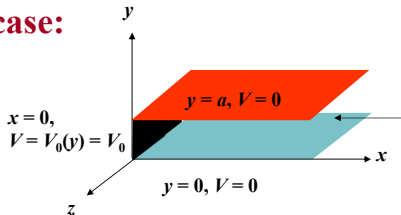
$$\text{OR } C_n = \frac{2}{a} \int_0^a V_0(y) \sin(n\pi y/a) dy$$

$$V(x, y) = \sum_{n=1}^{\infty} \left\{ \frac{2}{a} \int_0^a V_0(y) \sin\left(\frac{n\pi y}{a}\right) dy \right\} e^{-n\pi x/a} \sin(n\pi y/a)$$

This is the only solution to LE for this problem

Special Techniques in Electrostatics: Example 1

Special case:



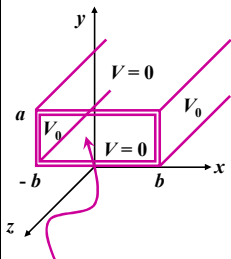
What is V here?

$$V(x, y) = \sum_{n=1}^{\infty} C_n e^{-n\pi x/a} \sin(n\pi y/a)$$

$$C_n = \frac{2}{a} \int_0^a V_0(y) \sin(n\pi y/a) dy = \frac{2V_0}{a} \int_0^a \sin(n\pi y/a) dy = \frac{2V_0}{a} \frac{a}{n\pi} \left[-\cos\left(\frac{n\pi y}{a}\right) \right]_0^a$$
$$= \frac{2V_0}{n\pi} (1 - \cos n\pi) = \begin{cases} 0 & n \text{ being even} \\ \frac{4V_0}{n\pi} & n \text{ being odd} \end{cases}$$

$$V(x, y) = \frac{4V_0}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n} e^{-n\pi x/a} \sin(n\pi y/a)$$

Special Techniques in Electrostatics: Example 2



What is V inside?

Symmetry

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

Define BC

- i. $V = 0$, at $y = 0$
- ii. $V = 0$, at $y = a$
- iii. $V = V_0$ at $x = b$
- iv. $V = V_0$ at $x = -b$

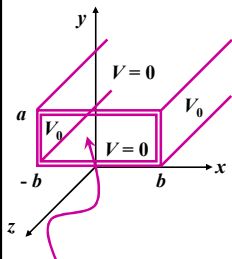
$$V(x, y) = (Ae^{kx} + Be^{-kx})(C \sin ky + D \cos ky)$$

$A \neq 0$, as there is no extension for x to infinity

However, there is symmetry w.r.t. x : $V(-x, y) = V(x, y) \rightarrow A = B$

$$A(e^{kx} + e^{-kx}) = 2A \cosh kx$$

Special Techniques in Electrostatics: Example 2



What is V inside?

Symmetry

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

Define BC

i. $V = 0$, at $y = 0$

ii. $V = 0$, at $y = a$

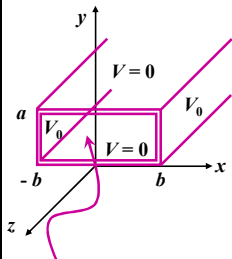
iii. $V = V_0$ at $x = b$

iv. $V = V_0$ at $x = -b$

$D = 0$

$$V(x, y) = \cosh kx (C \sin ky + D \cos ky)$$

Special Techniques in Electrostatics: Example 2



What is V inside?

Symmetry

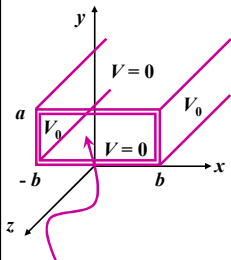
$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

Define BC

- i. $V = 0$, at $y = 0$
- ii. $V = 0$, at $y = a$
- iii. $V = V_0$ at $x = b$
- iv. $V = V_0$ at $x = -b$

$$V(x, y) = \cosh kx C \sin ky \quad k = n\pi / a$$

Special Techniques in Electrostatics: Example 2



What is V inside?

Symmetry

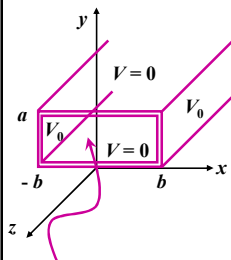
$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

Define BC

- i. $V = 0$, at $y = 0$
- ii. $V = 0$, at $y = a$
- iii. $V = V_0$ at $x = b$
- iv. $V = V_0$ at $x = -b$

$$V(x, y) = C \cosh(n\pi x / a) \sin(n\pi y / a) \quad k = n\pi / a$$

Special Techniques in Electrostatics: Example 2



What is V inside?

Symmetry

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

Define BC

- i. $V = 0$, at $y = 0$
- ii. $V = 0$, at $y = a$
- iii. $V = V_0$ at $x = b$
- iv. $V = V_0$ at $x = -b$

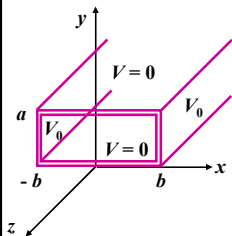
$$V(x, y) = C \cosh(n\pi x / a) \sin(n\pi y / a)$$

General Solution

$$V(x, y) = \sum_{n=1}^{\infty} C_n \cosh(n\pi x / a) \sin(n\pi y / a)$$

$$V(b, y) = \sum_{n=1}^{\infty} C_n \cosh(n\pi b / a) \sin(n\pi y / a) = V_0$$

Special Techniques in Electrostatics: Example 2



$$\begin{aligned}
 & \int_0^a \sum_{n=1}^{\infty} C_n \cosh(n\pi b/a) \sin(n\pi y/a) \sin(n'\pi y/a) dy \\
 &= \sum_{n=1}^{\infty} C_n \cosh(n\pi b/a) \int_0^a \sin(n\pi y/a) \sin(n'\pi y/a) dy \\
 &= \sum_{n=1}^{\infty} C_n \cosh(n\pi b/a) \frac{a}{2} \delta_{nn'} = V_0 \int_0^a \sin(n'\pi y/a) dy
 \end{aligned}$$

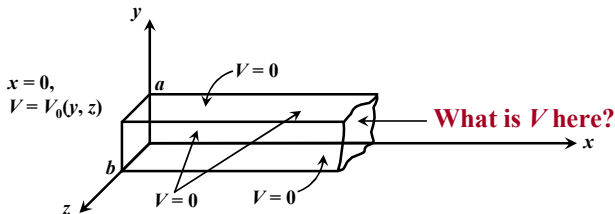
$$\longrightarrow C_n \cosh(n\pi b/a) = \frac{2V_0}{a} \int_0^a \sin(n'\pi y/a) dy = \begin{cases} 0 & n \text{ being even} \\ \frac{4V_0}{n\pi} & n \text{ being odd} \end{cases}$$

$$C_n = \frac{4V_0}{n\pi} \frac{1}{\cosh(n\pi b/a)} \quad n = 1, 3, 5, \dots$$

$$V(x, y) = \sum_{n=1}^{\infty} C_n \cosh(n\pi x/a) \sin(n\pi y/a)$$

$$V(x, y) = \frac{4V_0}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \frac{\cosh(n\pi x/a)}{\cosh(n\pi b/a)} \sin(n\pi y/a)$$

Special Techniques in Electrostatics: Example 3



1. Look for symmetry $V = V(x, y, z)$

2. DE: LE as there is no charge in the region of interest

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

3. Define BC

i. $V = 0$, at $y = 0$

iv. $V = 0$ at $z = b$

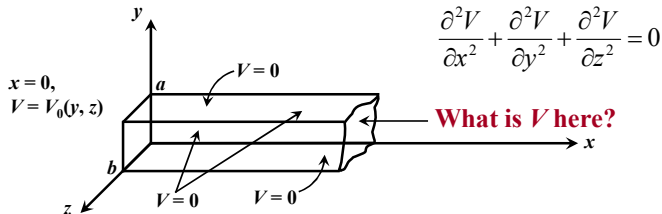
ii. $V = 0$, at $y = a$

v. $V \rightarrow 0$, at $x \rightarrow \infty$

iii. $V = 0$ at $z = 0$

vi. $V = V_0(y, z)$ at $x = 0$

Special Techniques in Electrostatics: Example 3



4. Variable separable method

$$V(x, y) = X(x)Y(y)Z(z)$$

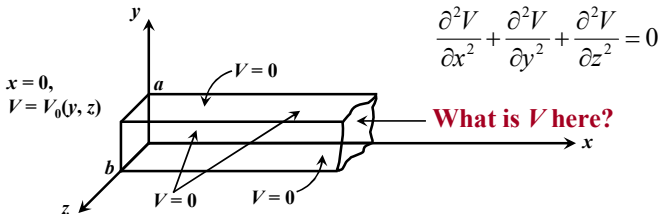
$$YZ \frac{d^2 X}{dx^2} + XZ \frac{d^2 Y}{dy^2} + XY \frac{d^2 Z}{dz^2} = 0$$

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = 0$$

$$f(x) + g(y) + h(z) = 0$$

$$C_1 + C_2 + C_3 = 0$$

Special Techniques in Electrostatics: Example 3



$$\longrightarrow \frac{1}{X} \frac{d^2 X}{dx^2} = C_1 \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = C_2 \quad \frac{1}{Z} \frac{d^2 Z}{dz^2} = C_3 \quad \text{PDE} \rightarrow \text{ODE}$$

As $V \rightarrow 0$, at $x \rightarrow \infty$, solution for x must be exponentially decaying

Along y and z , solution must be sinusoidal to meet BC

$\longrightarrow C_1$ must +ve, C_2 and C_3 must be -ve

$$C_2 = -k^2 \quad C_3 = -l^2 \quad C_1 = k^2 + l^2$$

Special Techniques in Electrostatics: Example 3

$$\frac{d^2 X}{dx^2} - (k^2 + l^2)X = 0 \quad \frac{d^2 Y}{dy^2} + k^2 Y = 0 \quad \frac{d^2 Z}{dz^2} + l^2 Z = 0$$

$$X(x) = Ae^{\sqrt{k^2 + l^2}x} + Be^{-\sqrt{k^2 + l^2}x}$$

$$Y(y) = C \sin ky + D \cos ky$$

$$Z(z) = E \sin lz + F \cos lz$$

Apply BC i. $V = 0$, at $y = 0$ $\longrightarrow D = 0$

iii. $V = 0$, at $z = 0$ $\longrightarrow F = 0$

v. $V \rightarrow 0$ as $x \rightarrow \infty$ $\longrightarrow A = 0$

ii $V = 0$ at $y = a$ $\longrightarrow k = n\pi / a \quad n = 1, 2, 3, \dots$

iv $V = 0$, at $z = b$ $\longrightarrow l = m\pi / b \quad m = 1, 2, 3, \dots$

$$V(x, y, z) = Ce^{-x\sqrt{k^2 + l^2}} \sin(n\pi y / a) \sin(m\pi z / b)$$

Special Techniques in Electrostatics: Example 3

$$\frac{d^2 X}{dx^2} - (k^2 + l^2)X = 0 \quad \frac{d^2 Y}{dy^2} + k^2 Y = 0 \quad \frac{d^2 Z}{dz^2} + l^2 Z = 0$$

$$X(x) = Ae^{\sqrt{k^2 + l^2}x} + Be^{-\sqrt{k^2 + l^2}x}$$

$$Y(y) = C \sin ky + D \cos ky$$

$$Z(z) = E \sin lz + F \cos lz$$

Apply BC i. $V = 0$, at $y = 0$ $\longrightarrow D = 0$

iii. $V = 0$, at $z = 0$ $\longrightarrow F = 0$

v. $V \rightarrow 0$ as $x \rightarrow \infty$ $\longrightarrow A = 0$

ii $V = 0$ at $y = a$ $\longrightarrow k = n\pi / a \quad n = 1, 2, 3, \dots$

iv $V = 0$, at $z = b$ $\longrightarrow l = m\pi / b \quad m = 1, 2, 3, \dots$

General Solution

$$V(x, y, z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} e^{-x\pi\sqrt{(n/a)^2 + (m/b)^2}} \sin(n\pi y / a) \sin(m\pi z / b)$$

Special Techniques in Electrostatics: Example 3

General Solution

$$V(x, y, z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} e^{-x\pi\sqrt{(n/a)^2 + (m/b)^2}} \sin(n\pi y/a) \sin(m\pi z/b)$$

Apply BC vi. $V = V_0(y, z)$, at $x = 0$

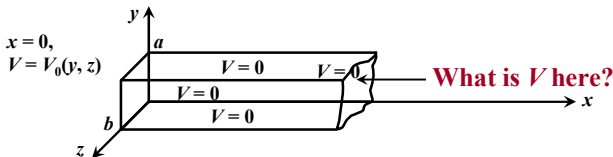
$$V(0, y, z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \sin(n\pi y/a) \sin(m\pi z/b) = V_0(y, z)$$

To get C_{nm} , use Fourier Trick by multiplying $\sin(n'\pi y/a)$ and $\sin(m'\pi z/b)$ and integrate from $0 - a$ and $0 - b$

$$\begin{aligned} & \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \int_0^a \sin(n\pi y/a) \sin(n'\pi y/a) dy \int_0^b \sin(m\pi z/b) \sin(m'\pi z/b) dz \\ &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \frac{a}{2} \delta_{nn'} \frac{b}{2} \delta_{mm'} = \int_0^a \int_0^b V_0(y, z) \sin(n'\pi y/a) \sin(m'\pi z/b) dy dz \end{aligned}$$

$$\longrightarrow C_{n'm'} = \frac{4}{ab} \int_0^a \int_0^b V_0(y, z) \sin(n'\pi y/a) \sin(m'\pi z/b) dy dz$$

Special Techniques in Electrostatics: Example 3



Unique Solution

$$V(x, y, z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} e^{-x\pi\sqrt{(n/a)^2 + (m/b)^2}} \sin(n\pi y / a) \sin(n\pi z / b)$$

$$C_{n'm'} = \frac{4}{ab} \int_0^a \int_0^b V_0(y, z) \sin(n'\pi y / a) \sin(m'\pi z / b) dy dz$$

Special case: $V_0(y, z) = V_0$

$$C_{n'm'} = \frac{4V_0}{ab} \int_0^a \sin(n'\pi y / a) dy \int_0^b \sin(m'\pi z / b) dz = \begin{cases} 0 & n, m \text{ being even} \\ \frac{16V_0}{nm\pi^2} & n, m \text{ being odd} \end{cases}$$

$$V(x, y, z) = \frac{16V_0}{\pi^2} \sum_{n=1,3,5,\dots}^{\infty} \sum_{m=1,3,5,\dots}^{\infty} \frac{1}{nm} e^{-x\pi\sqrt{(n/a)^2 + (m/b)^2}} \sin(n\pi y / a) \sin(n\pi z / b)$$

